Pour cet exemple, je prends $N_a = N_b = 12{,}65$ $\left(\frac{N_a + N_b}{2}\right)$

<u>Mélange idéal :</u>

$$\Delta G = k_B T \left[\frac{\phi}{N_a} \ln \phi + \frac{(1-\phi)}{N_b} \ln (1-\phi) \right]$$

<u>Mélange réel :</u>

on ajoute $\chi_{AB} = \frac{V_0}{k_B T} (\delta_A - \delta_B)^2$

$V_0 = 100 \text{ Å}^3$ $\quad \delta_A = 17{,}9 \text{ MPa}^{1/2}$

$\delta_B = 15{,}7 \text{ MPa}^{1/2}$

$$\chi_{AB} = \frac{1}{T} \times \frac{100 \times 10^{-30}}{8{,}314} \times 6{,}02 \times 10^{23} \left(17{,}9 \times 10^3 - 15{,}7 \times 10^3\right)^2$$

$$\chi_{AB} = \frac{35}{T} \qquad (B = 35)$$

$$\Delta G = k_B T \left[\frac{\phi}{N_a} \ln \phi + \frac{(1-\phi)}{N_b} \ln (1-\phi) + \chi_{AB} \phi (1-\phi) \right]$$

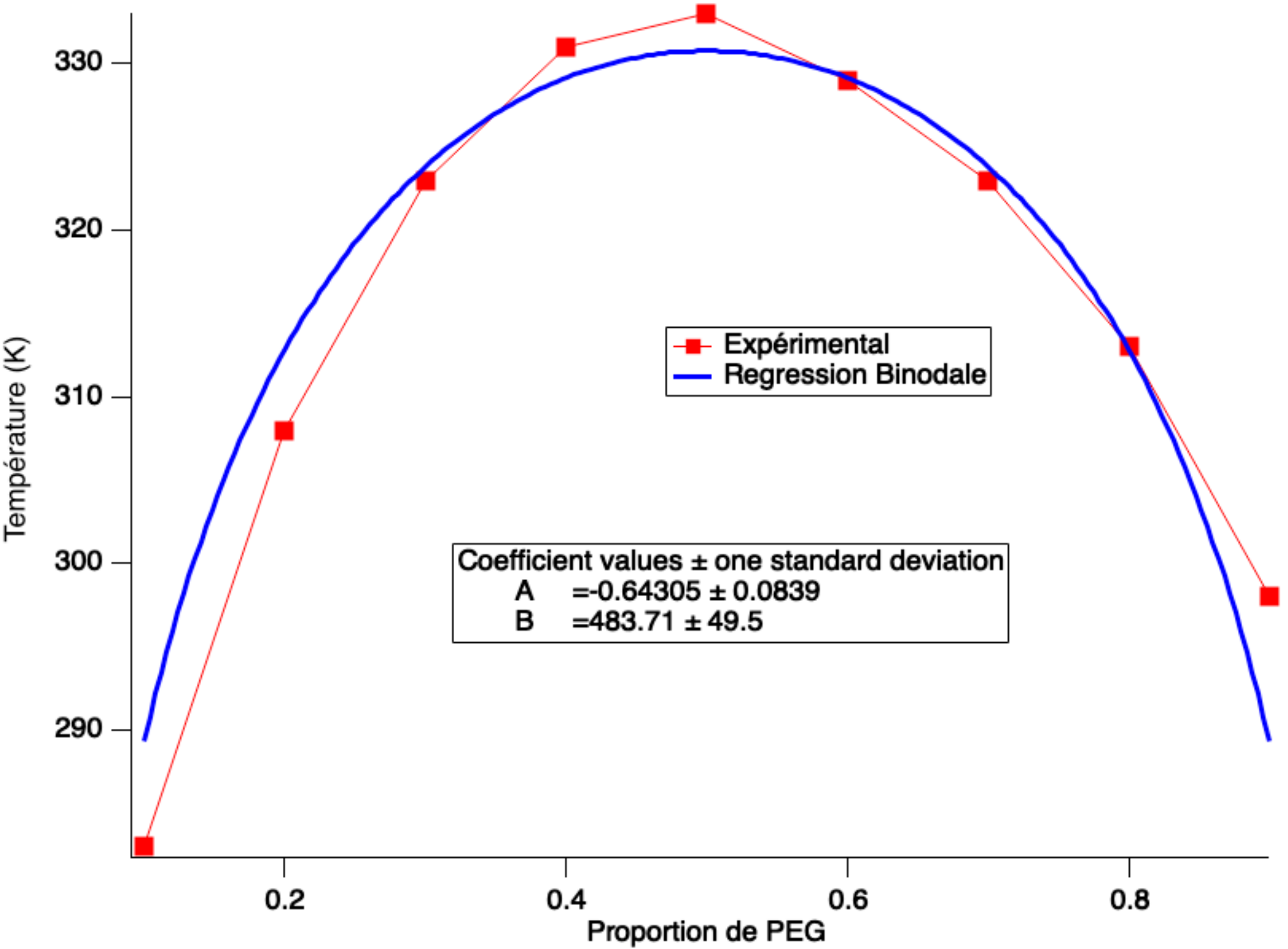
<u>Température Binodale :</u>

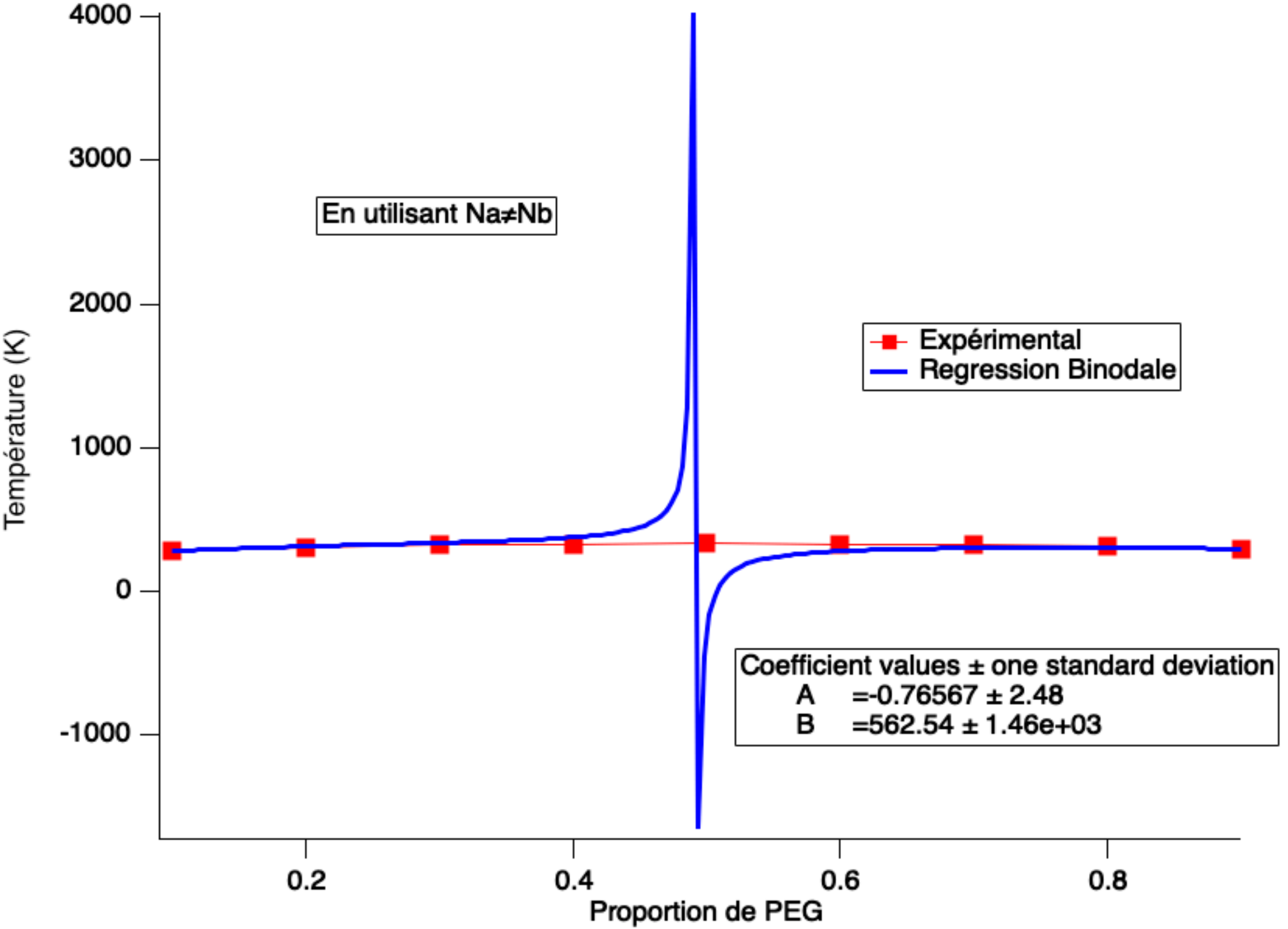
$$T_B = \frac{B}{[\ldots] - A}$$

Dans cet exemple, en utilisant Igor Pro pour faire la regression,

~~A = -0,26~~
~~B = 143~~
~~en °C~~

A = -0,684
B = ~~840~~ 484
en kelvin


330
320
310
300
290
Température (K)
0.2
0.4
0.6
0.8
Proportion de PEG
Expérimental
Regression Binodale
Coefficient values ± one standard deviation
A =-0.64305 ± 0.0839
B =483.71 ± 49.5


En utilisant Na≠Nb
Expérimental
Regression Binodale
Coefficient values ± one standard deviation
A =-0.76567 ± 2.48
B =562.54 ± 1.46e+03
Température (K)
Proportion de PEG
4000
3000
2000
1000
0
-1000
0.2
0.4
0.6
0.8